

Representation and Storage of Digital Information

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Table of Contents i

Introduction

Digital Representation

Data Hierarchy

Unstructured Data

Semi-Structured Data

Multi-Format Comparison

Binary Serialization

Tools and Validation

Summary

Introduction

The Role of Data in Engineering I

As engineers, we deal with data in almost every task.

- **Sensor Readings:** Temperature, pressure, location.
- **Configurations:** How a system should behave.
- **Logs:** What happened in the past.
- **Communication:** Sending messages between services.
- **Storage:** Saving state for later use.

The Role of Data in Engineering II

The quality of our systems depends on how we represent this data.

- **Efficiency:** Minimizing storage and bandwidth.
- **Reliability:** Ensuring data is not corrupted.
- **Interoperability:** Allowing different systems to “talk” to each other.
- **Maintainability:** Making it easy for humans to understand and change.

Data vs. Information I

It is important to distinguish between these two concepts.

- **Data:** Raw facts, symbols, or signals without context (e.g., [25, 26, 25, 24]).
- **Information:** Data that has been processed, organized, or structured to be meaningful (e.g., "The average temperature in the room was 25°C").

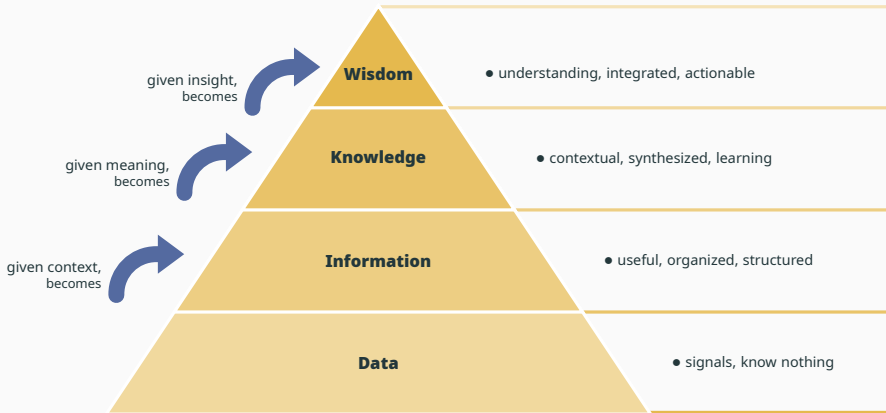
Data vs. Information II

- Data is the input; Information is the output.
- Information requires **context**.
- Without a schema or metadata, data is just a collection of bits.
- In this class, we focus on how to transform raw bits into structured information.

The DIKW Pyramid

A model to represent the structural and functional relationships between data and wisdom.

- **Data:** The foundation (symbols).
- **Information:** Linked data (answers who, what, where, when).
- **Knowledge:** Applied information (answers how).
- **Wisdom:** Evaluated knowledge (answers why).



Digital Representation

At the lowest level, all digital information is binary.

- **Bit:** The smallest unit of information (0 or 1).
- **Byte:** A group of 8 bits.
- **Capacity:** A byte can represent $2^8 = 256$ different values.
- **Hexadecimal:** A more human-friendly way to write bytes (e.g., 0xFF).

How do we represent numbers?

- **Integers:** Fixed-point representation (e.g., 2's complement).
- **Floats:** Floating-point representation (IEEE 754).
- **Endianness:** The order of bytes in memory (Big-Endian vs. Little-Endian).
- **Network Order:** Usually Big-Endian.

Character Encoding I: ASCII

How do we represent text?

- **ASCII (1963):** American Standard Code for Information Interchange.
- **Limit:** 7 bits (128 characters).
- **Coverage:** English alphabet, numbers, and basic symbols.
- **Problem:** No support for accented characters (á, ç), Greek letters, or Emojis.

Character Encoding II: Unicode

The solution to the “Babel” of encodings.

- **Goal:** A single standard for every character in every language.
- **Code Points:** Each character is assigned a unique number (e.g., U+0041 for 'A').
- **Size:** Supports over 1 million characters.

Character Encoding III: UTF-8

The most popular encoding for the web and modern systems.

- **Variable Length:** Uses 1 to 4 bytes per character.
- **Backward Compatible:** The first 128 characters are identical to ASCII.
- **Efficiency:** English text takes 1 byte per char; more complex scripts take more.
- **Standard:** Should be the default choice for all engineering projects.

Data Hierarchy

Classification of Data I

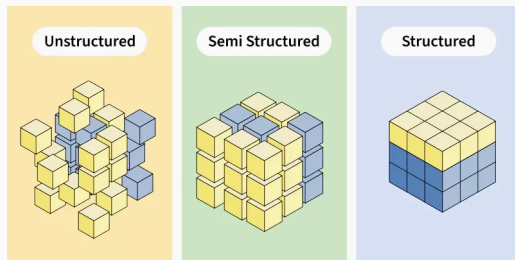
Data is generally classified into three categories based on its structure:

1. **Unstructured Data:** No predefined format.
2. **Semi-Structured Data:** Has some organizational properties but no strict schema.
3. **Structured Data:** Follows a strict, predefined model (usually tabular).

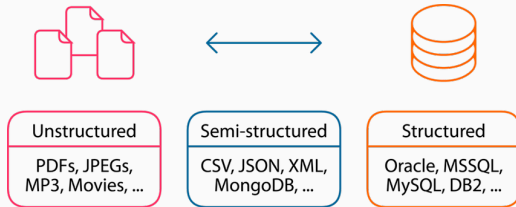
Understanding how data is organized is the first step in the pipeline.

- **Unstructured:** Text, Binary, Media.
- **Semi-Structured:** Hierarchical trees or Key-Value pairs (JSON, XML).
- **Structured:** Tabular (Rows and Columns).

Illustration of Structure



Data Structure Spectrum



Examples of File Formats

Unstructured Data

How is the data stored on disk?

- **Text Files:** Human-readable sequences of characters. Debuggable and portable.
- **Binary Files:** Sequences of bytes for machine reading. Compact and fast.
- **Magic Numbers:** First bytes identifying the format (e.g., 0x89 0x50 0x4E 0x47 for PNG).

Processing Unstructured Data: Basic Tools

Before using advanced tools, we use basic Unix utilities to explore text:

- `head / tail`: View the beginning or end of a file.
- `wc`: Count lines, words, and characters.
- `file`: Identify the type of data.
- `grep`: Search for patterns.

Regular Expressions (Regex) I

To process unstructured text, we need a way to describe patterns.

- **Literal:** abc matches "abc".
- **Wildcard:** . matches any character.
- **Quantifiers:** * (0+), + (1+), ? (0 or 1).
- **Character Classes:** [a-z], \d (digit), \w (word), \s (whitespace).

Regular Expressions (Regex) II

- **Anchors:** ^ (start), \$ (end).
- **Example (Email):** `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`

Logs are “mostly” unstructured text.

- **awk:** Process columns of text.
 - Example: `awk '$9 == 404 {print $7}' access.log`
- **sed:** Stream editor for transformation.
 - Example: `sed 's/old/new/g' file.txt`

Representing visual and complex information.

- **Raster:** Grid of pixels (PNG, JPEG). Lossy vs. Lossless.
- **Vector:** Mathematical paths (SVG, PDF). Infinitely scalable.
- **PDF:** Container for text, fonts, and graphics. Includes **Metadata** (Author, Date).

Semi-Structured Data

Data that uses “tags” or “markers” to separate semantic elements.

- **Flexibility:** Represents hierarchy and nesting.
- **Formats:** CSV, XML, JSON, YAML.

CSV (Comma Separated Values)

The simplest way to exchange tabular data as text.

- **Concept:** Each line is a record; fields are separated by a delimiter (,, ;, or tab).
- **Pros:** Universally supported, lightweight.
- **Cons:** No types, no nesting, quoting issues (fields with delimiters must be "quoted").

XML (eXtensible Markup Language)

An older, robust standard using nested tags.

- **Pros:** Supports complex schemas (XSD) and namespaces.
- **Cons:** Verbose, tag bloat, slower to parse.
- **Example:** `<user id="1"><name>mario</name></user>`

JSON (JavaScript Object Notation)

The king of modern web communication (APIs).

- **Types:** Objects {}, Arrays [], Strings, Numbers, Booleans, Null.
- **Pros:** Smaller than XML, native to Python Dictionaries and Lists.
- **Validation:** Use **JSON Schema** to define rules.

YAML (YAML Ain't Markup Language)

The standard for configuration and DevOps.

- **Syntax:** Relies on indentation instead of brackets.
- **Features:** Supports comments #, anchors, and multi-line strings.
- **Usage:** Docker, Kubernetes, GitHub Actions.

Multi-Format Comparison

Employee Record: CSV

No native nesting. “Skills” list requires a custom separator (e.g., pipe |).

```
id,name,skills,active
1,"Jane Doe","Python|SQL",true
2,"Bob Smith","Java|C++",false
```

Employee Record: XML

```
<employees>
  <employee id="1">
    <name>Jane Doe</name>
    <skills><skill>Python</skill><skill>SQL</skill></skills>
    <active>true</active>
  </employee>
</employees>
```

Employee Record: JSON

```
{"employees": [{  
    "id": 1, "name": "Jane Doe",  
    "skills": ["Python", "SQL"], "active": true  
}]  
}
```

Employee Record: YAML

```
employees:  
  - id: 1  
    name: Jane Doe  
    skills: [Python, SQL]  
    active: true
```

Binary Serialization

When performance is more important than human readability.

- **BSON:** Binary JSON (used in MongoDB). Supports more types (e.g., Date).
- **MessagePack:** Small, fast binary serialization.
- **Parquet:** Columnar storage format. Efficient for big data analytics.

Google's serialization format.

1. **Define:** Structure in a `.proto` file.
2. **Compile:** Generate code for your language.
3. **Serialize:** Highly compact binary data for transmission.

Tools and Validation

Working with Data: jq and yq

Command-line tools for processing JSON and YAML.

- **jq:** `cat data.json | jq '.user'`
- **yq:** `yq eval '.port = 8080' config.yml`

Data Validation with Pydantic

Pydantic is the modern way to validate data in Python.

```
from pydantic import BaseModel, EmailStr

class User(BaseModel):
    id: int
    name: str
    email: EmailStr

# Validates and converts types automatically
u = User(id="1", name="Mario", email="mario@ua.pt")
```

Summary

Summary

- **Categories:** Unstructured (logs), Semi-structured (JSON/YAML), Structured (CSV).
- **Fundamentals:** Understand bits, bytes, and character encodings (UTF-8).
- **Processing:** Use Regex, awk, and sed for text; jq/yq for formats.
- **Choice:** Use JSON for APIs, YAML for config, and CSV/Parquet for bulk data.
- **Validation:** Always validate data (e.g., Pydantic) to ensure system integrity.